

TY-MDM | Generative AI

Sample Question Paper 1

Total Marks: 30

Q1. Attempt the following (Compulsory)

(a) A small online clothing business uses a Generative AI tool to create product descriptions automatically. Apply the concept of Generative AI to demonstrate how it can improve operational efficiency and reduce workload in this business. **[4 Marks]**

(b) A hospital uses an AI system to assist doctors in diagnosing diseases. The system provides accurate results but does not explain how decisions are made, leading to distrust among doctors. Analyze the scenario and explain the significance of Responsible AI in this context. **[4 Marks]**

Q2. Attempt ANY 2 out of 3 (*Attempt any 2 out of the following 3 questions*)

(a) A fintech startup develops a trading system using GAN-generated synthetic stock data due to limited real data. Apply how synthetic data improves model performance and robustness. **[4 Marks]**

(b) A pharmaceutical startup has a dataset of known drug compounds. Apply a generative AI technique to generate new candidate molecules for further testing. **[4 Marks]**

(c) A startup uses AI tools to generate social media posts daily. Analyze how Generative AI influences marketing strategies in terms of personalization, audience targeting, and content effectiveness. **[4 Marks]**

Q3.

(A) A biomedical lab uses GANs to generate synthetic MRI brain scan images for detecting tumors due to limited real patient data. Illustrate the role of GAN-generated MRI brain scans in improving the training dataset for tumor detection models. **(Compulsory) [4 Marks]**

(B) A loan approval AI system consistently rejects applications from certain neighborhoods due to historical financial data patterns. Analyze how such bias can affect economic equality and social mobility. **[3 Marks]**

OR

(C) A mobile application collects user data to provide personalized recommendations, but users are unaware of how their personal information is stored and used. A data breach later exposes sensitive information. Examine the

privacy risks associated with this AI system and their potential consequences. **[3 Marks]**

Q4. Attempt the following

(a) A company uses Large Language Models for customer service automation. Analyze scalability benefits. **[4 Marks]**

(b) Identify key components of Responsible AI. **[3 Marks]**

Sample Question Paper 2

Total Marks: 30

Q1. Attempt the following (Compulsory)

- (a) A bank uses AI-generated reports for financial analysis. Illustrate how Generative AI improves decision-making efficiency. **[4 Marks]**
- (b) A company automates invoice processing using AI. Apply how this reduces errors and improves accuracy. **[4 Marks]**

Q2. Attempt ANY 2 out of 3 (Attempt any 2 out of the following 3 questions)

- (a) A climate research intern is given limited temperature data for a region. Demonstrate the use of generative AI to create additional synthetic climate records. **[4 Marks]**
- (b) A global research team works across different time zones and is willing to use digital platforms to collaborate on research projects and report writing.
(a) Recommend suitable generative AI tools to assist in creating and editing collaborative research reports. (b) Illustrate the importance of collaborative work in global research teams using generative AI tools. **[4 Marks]**
- (c) A marketing company uses Generative AI to automatically create advertisements based on historical customer data. However, the training data contains inherent biases. Analyze how bias in Generative AI-generated advertisements can impact marketing outcomes such as audience targeting, brand perception, and campaign effectiveness. Support your answer with a suitable example. **[4 Marks]**

Q3.

- (A) A multinational company develops an AI-based hiring system and adopts a single ethical guideline focused only on accuracy and efficiency. Over time, the system begins to show unfair hiring patterns across different demographic groups. Evaluate the suitability of the ethical framework used and assess how a more comprehensive framework could improve outcomes. **(Compulsory)**
[4 Marks]

- (B) Illustrate transparency in AI with a simple example. **[3 Marks]**

OR

- (B) Apply techniques to reduce bias in datasets. **[3 Marks]**

Q4. Attempt the following

(a) A healthcare organization deploys an AI system to assist doctors in diagnosing diseases, but final decisions are always reviewed by medical experts before implementation. Illustrate how human-in-the-loop mechanisms function in this system and their importance. **[4 Marks]**

(b) List ethical principles applicable to AI systems. **[3 Marks]**

Sample Question Paper 3

Total Marks: 30

Q1. Attempt the following (Compulsory)

- (a) A software company uses AI to generate code snippets. Apply the concept of Generative AI to demonstrate how it enhances software development productivity. **[4 Marks]**

- (b) A trading platform leverages Generative AI to create simulated market scenarios such as bullish, bearish, and highly volatile conditions. (a) Apply how Generative AI can be used to generate realistic synthetic market data and trading environments to enhance and optimize trading strategies. (b) Support your answer with an example demonstrating improved trading decisions. **[4 Marks]**

Q2. Attempt ANY 2 out of 3 . (*Attempt any 2 out of the following 3 questions*)

- (a) A biotech company is working on vaccine development. Apply generative AI to generate new protein structures using existing datasets. **[4 Marks]**

- (b) A bank deploys an LLM-based chatbot for handling customer queries. Apply how this improves service efficiency and user experience. **[4 Marks]**

- (c) A social media company uses AI algorithms to recommend content. Users complain about misinformation, while the company focuses on engagement metrics. Analyze the roles of platform owners, developers, users, and regulators in managing such issues. **[4 Marks]**

Q3.

- (A) A startup plans to build an AI system for loan approval that must ensure fairness, transparency, and accountability from the beginning. Design a structured framework for developing an unbiased AI system, considering all stages from data collection to deployment. **(Compulsory) [4 Marks]**

- (B) Summarize the concept of fairness in AI systems. **[3 Marks]**

OR

- (B) Demonstrate preprocessing methods for unbiased AI models. **[3 Marks]**

Q4. Attempt the following

(a) A global logistics firm wants to predict and handle disruptions such as port delays and sudden demand spikes. Using Generative AI, analyze how synthetic scenario generation can help simulate disruptions and improve supply chain resilience. Explain with an example. **[4 Marks]**

(b) Outline basic privacy concerns in AI applications. **[3 Marks]**

Sample Question Paper 4

Total Marks: 30

Q1. Attempt the following (Compulsory)

- (a) A teacher uses AI to generate quizzes for students. Apply the concept of Generative AI to demonstrate how it can be used to create personalized learning assessments. **[4 Marks]**
- (b) A financial institution uses an AI-based system to automatically approve or reject loan applications without human intervention. Some applicants are denied without clear explanations. Analyze the ethical concerns arising from automated decision-making in this scenario. **[4 Marks]**

Q2. Attempt ANY 2 out of 3 (Attempt any 2 out of the following 3 questions)

- (a) A material science lab wants variations of a chemical compound. Apply generative AI to create multiple compound structures. **[4 Marks]**
- (b) A firm uses Generative AI to automate the generation of financial reports from business data. Apply the concept of Generative AI to explain its use in automating financial report generation and improving productivity in the organization. **[4 Marks]**
- (c) A university integrates AI tools for research paper drafting. Analyze academic integrity issues. **[4 Marks]**

Q3.

- (A) A well-known tech company deploys an AI-powered facial recognition system that later shows high error rates for certain communities, leading to public criticism and regulatory action. Analyze this real-world failure and highlight the role of bias in causing these issues along with its consequences. **(Compulsory) [4 Marks]**

- (B) Recognize examples of algorithmic bias in AI applications. **[3 Marks]**

OR

- (B) Differentiate fairness and equity in AI systems. **[3 Marks]**

Q4. Attempt the following

- (a) A multidisciplinary research project requires collaboration among experts from different domains. Using Generative AI, suggest appropriate tools that can facilitate interdisciplinary research activities such as knowledge integration, content creation, and collaborative communication. Explain your answer with suitable examples. **[4 Marks]**
- (b) Analyze responsibilities of developers in ethical AI practices. **[3 Marks]**

Sample Question Paper 5

Total Marks: 30

Q1. Attempt the following (Compulsory)

- (a) A travel agency uses AI to generate personalized travel plans. Apply the concept of Generative AI to show how it can enhance customer experience through personalization. **[4 Marks]**
- (b) An organization uses AI to automatically generate financial summaries for stakeholders based on large datasets. (a) Analyze the challenges related to accuracy in AI-generated financial summaries for stakeholders. (b) Examine the issues of accountability and responsibility when financial summaries are generated using AI systems. **[4 Marks]**

Q2. Attempt ANY 2 out of 3 (Attempt any 2 out of the following 3 questions)

- (a) Public health officials want to study disease spread. Apply generative AI to simulate infection scenarios under different conditions. **[4 Marks]**
- (b) A marketing agency generates personalized ads using customer data. Apply how Generative AI improves creativity and personalization. **[4 Marks]**
- (c) A government uses AI tools to generate policy drafts. Examine benefits and risks. **[4 Marks]**

Q3.

- (A) An AI model used in healthcare shows biased predictions due to imbalanced training data. The development team considers multiple bias mitigation approaches at different stages of the pipeline. Evaluate various bias mitigation techniques and justify the most effective approach for this scenario with suitable examples. (Compulsory) **[4 Marks]**
- (B) Analyze ethical challenges in Generative AI systems. **[3 Marks]**

OR

- (C) Interpret regulatory roles in governing AI systems. **[3 Marks]**

Q4. Attempt the following

- (a) A healthcare organization wants to personalize drug development based on patient-specific genetic information. Using Generative AI, recommend how generative models can be used to design personalized drugs and simulate their outcomes before clinical trials. Justify your approach. **[4 Marks]**
 - (b) Examine sources through which bias enters AI models. **[3 Marks]**
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